ModelIt!

Teacher's Guide

Why Do Some Countries Run Out of Water?

Modeling Water Stress from Rainfall, Population, and Consumption

Quick Reference

Grade Level	Grade 06L2
NGSS Standard	MS-ESS3-1, MS-ESS3-4
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-ESS3-1, MS-ESS3-4

Construct a scientific explanation based on evidence for how the uneven distributions of Earth's mineral, energy, and groundwater resources are the result of past and current geoscience processes. Construct an argument supported by evidence for how increases in human population and per-capita consumption of natural resources impact Earth's systems.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model showing how rainfall patterns and population density interact through aquifer recharge and water demand to produce a Water Stress Index indicating sustainability or crisis.
Disciplinary Core Idea	**ESS3.A Natural Resources / ESS3.C Human Impacts on Earth Systems** — Humans depend on Earth's land, ocean, atmosphere, and biosphere for freshwater resources. Increases in human population and per-capita consumption of natural resources impact Earth's groundwater systems.
Crosscutting Concept	**Stability and Change** — Students investigate how the balance between water supply and demand can shift from stable sustainability to unstable crisis through changes in rainfall, population, or consumption patterns.

CAST Testing Alignment

MULTIPLE CHOICE:

City A and City B both receive 500 mm of annual rainfall. City A has 100,000 residents and City B has 1,000,000 residents. Based on the water stress model, which city is more likely to face a water crisis, and what is the primary factor causing the difference?

CONSTRUCTED RESPONSE:

Using your water stress model, explain why Cape Town, South Africa nearly ran out of water in 2018 even though it receives moderate annual rainfall. Reference rainfall patterns, population growth, aquifer recharge rates, and the Water Stress Index in your explanation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Water is the most essential resource for human civilization, yet billions of people face water scarcity. Understanding why requires modeling the complex balance between natural water supply and human water demand, a balance that is shifting in dangerous directions around the world.

Uneven Distribution of Freshwater

Only about 2.5% of Earth's water is freshwater, and most of that is locked in ice caps and glaciers. The freshwater available for human use (rivers, lakes, accessible groundwater) is unevenly distributed by geography and climate. Some regions receive heavy rainfall while others are arid. This natural inequality means water stress is fundamentally a geographic problem, not a global shortage.

Aquifers: Underground Water Banks

Aquifers are layers of permeable rock or sediment that store groundwater. They are recharged when rainfall filters through the soil, a process that can take months to thousands of years depending on the aquifer depth and geology. Humans extract water from aquifers using wells and pumps. When extraction exceeds recharge (overdraft), the aquifer level drops, wells go dry, and land can permanently compact (subsidence), destroying storage capacity forever.

Population Growth and Water Demand

Global water demand has increased six-fold over the past century, driven primarily by population growth, agricultural expansion, and industrial development. Agriculture alone accounts for roughly 70% of freshwater withdrawal. As populations grow in water-stressed regions, the gap between demand and supply widens. Per-capita consumption also matters: an average American uses about 300 liters per day, while a person in sub-Saharan Africa may use only 20 liters.

Tipping Points and Water Crisis

Water systems can shift from stressed to crisis rapidly. Cape Town nearly reached 'Day Zero' in 2018, when taps would be shut off. The Aral Sea, once the world's fourth-largest lake, has nearly disappeared due to irrigation diversion. The Ogallala Aquifer in the US Great Plains is being depleted 3-10 times faster than it recharges. These examples show that water systems have tipping points beyond which recovery becomes extremely difficult or impossible.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify rainfall pattern and population density as the two external components, and aquifer recharge, water demand, and Water Stress Index as the three internal responses of the water stress system.

E **Establish Relationships**
Students determine that rainfall positively drives aquifer recharge, population density positively drives water demand, aquifer recharge negatively reduces water stress (more recharge means less stress), and water demand positively increases the Water Stress Index.

V **Visualize & Model**
Students build a model showing how rainfall feeds aquifer recharge while population drives demand, with the Water Stress Index reflecting the balance between supply and demand.

E **Evaluate Outcomes**
Students run sustainable balance, growing demand, and drought crisis scenarios to observe how the Water Stress Index responds to changes in either supply or demand.

R **Revise & Extend**
Students add climate change effects or agricultural water use to explore how additional factors compound water stress beyond basic supply and demand.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Earth has 326 million trillion gallons of water. That is a 326 followed by 18 zeros. So how is it possible that Cape Town almost turned off all its taps in 2018?"*

Watch For

Show a photo of Cape Town's empty reservoirs during the crisis. Let students react to the contrast between Earth's total water and a city's shortage.

Slide 2: Learning Objectives

3 min

Say: *"Today we are modeling why some countries face water crises while others have plenty. The answer is about balance, not total supply."*

Watch For

Have students read objectives. Pre-teach 'aquifer,' 'recharge rate,' and 'Water Stress Index.'

Slide 3: Big Question

3 min

Say: *"If the water cycle recycles all water and Earth never loses water, how can any place run out?"*

Watch For

Think-pair-share: Where does your tap water come from? What would happen if no rain fell for a year?

Slide 4: LEVER Framework

2 min

Say: *"Water stress is about the BALANCE between how fast nature refills the supply and how fast humans drain it. Our model captures this balance."*

Watch For

Draw a simple balance scale: rainfall and recharge on one side, population and demand on the other. Water Stress Index is the needle.

Slide 5: Activity 1: Components

8 min

Say: *"Rainfall and population are givens that we cannot control in the model. But recharge, demand, and the stress index respond to those inputs."*

Watch For

Guide sorting. Discuss why rainfall pattern is external (weather) and population density is external (demographics).

Slide 6: Activity 2: Connections

8 min

Say: *"Here is the critical insight: aquifer recharge is SLOW, measured in months to centuries. Water demand is FAST, measured in daily consumption. That time mismatch is what creates water stress."*

Watch For

Trace both paths: rainfall to recharge to supply, and population to demand. Show how the stress index compares the two.

Slide 7: Activity 3: Simulation

10 min

Say: *"What happens when a city triples in size but the rain stays the same? Let's model it."*

Watch For

Students predict outcomes. Run all three scenarios. Watch the Water Stress Index cross from green to yellow to red.

Slide 8: Discoveries

5 min

Say: *"Countries do not run out of water because Earth is drying up. They run out because they are using it faster than nature can replace it, and that is a choice we can change."*

Watch For

Show real-world examples: Ogallala Aquifer depletion, Aral Sea disappearance, Singapore's water recycling success.

Slide 9: STEM Challenge

12 min

Say: *"Your city is tripling in size and rainfall is dropping. Design a 50-year water plan that keeps the stress index below crisis."*

Watch For

Provide city data sheets with population projections and rainfall forecasts. Teams design management plans balancing conservation, infrastructure, and allocation.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Rainfall Pattern, Population Density
- **INTERNAL:** Aquifer Recharge, Water Demand, Water Stress Index

Relationships

- **Rainfall Pattern to Aquifer Recharge:** POSITIVE (+) — More rainfall provides more water that can infiltrate through soil and rock to replenish groundwater stores. Less rainfall means slower recharge.
- **Population Density to Water Demand:** POSITIVE (+) — More people in an area means more total water consumption for drinking, cooking, sanitation, agriculture, and industry, increasing overall demand.
- **Aquifer Recharge to Water Stress Index:** NEGATIVE (-) — Higher aquifer recharge means more water is being replenished, which reduces the gap between supply and demand and lowers the Water Stress Index.
- **Water Demand to Water Stress Index:** POSITIVE (+) — Higher water demand increases the gap between what humans need and what the environment can supply, raising the Water Stress Index toward crisis levels.
- **Rainfall Pattern to Water Stress Index (indirect):** NEGATIVE (-) — This indirect path shows how rainfall ultimately reduces water stress by feeding aquifer recharge, which increases available supply relative to demand.

Key Discoveries

- Water stress is not about total water on Earth but about the balance between local supply and local demand
- Aquifer recharge can take years to centuries, while human extraction can drain aquifers in decades
- Population growth increases water demand even when per-capita consumption stays constant
- Rainfall patterns are uneven across the globe, creating natural water-rich and water-poor regions
- The Water Stress Index can tip from sustainable to crisis when either supply drops or demand rises past a threshold

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Earth is running out of water"	Earth has the same amount of water it has always had. Water is not created or destroyed in the water cycle, only moved and transformed. The problem is distribution: freshwater is unevenly distributed, and humans are consuming it faster than nature replenishes it in certain locations. It is a local balance problem, not a global shortage.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Rainfall alone determines water stress"	Rainfall is only the supply side of the equation. Water stress depends on the BALANCE between supply and demand. A region with low rainfall but also low population may have no water stress, while a region with moderate rainfall but massive population can be in crisis. Both supply and demand must be modeled together.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Groundwater is unlimited because it refills from rain"	While aquifers do recharge from rainfall, the process can take years to thousands of years depending on depth and geology. Humans can extract water from an aquifer with a pump in minutes. This time mismatch means that aquifers can be depleted far faster than they recharge, effectively mining fossil water that took millennia to accumulate.